Supporting Information

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SI Text

Bounds on *q* Value. Consider the Clinton–Gore poll as an example with a pair of questions C and G. Define $p(C_yG_y)$ as the probability of the sequence for saying yes to question C and then yes to question G when C was asked first and G was asked second; define $p(G_yC_n)$ as the probability of the sequence for saying yes to question G and then no to question C when G was asked first and C was asked second; similar definitions are used for the other sequences. The context effect table contains four context effects, which can be defined as shown in Table S1.

The context effects must sum to zero because $(x+y+z+w) = p(G_yC_y) - p(C_yG_y) + p(G_nC_y) - p(C_yG_n) + p(G_yC_n) - p(C_nG_y) + p(G_nC_n) - p(C_nG_n) = 1 - 1 = 0$ so that $(x+w)+(y+z)=0 \rightarrow (y+z)=-(x+w)=q$.

The set of all context effects forms a 3D pyramid defined by all x, y, z, w such that $x+w+y+z=0$, $1-p(C_yG_y) \geq x \geq -p(C_yG_y)$, $1-p(C_yG_n) \geq y \geq -p(C_yG_n)$, $1-p(C_nG_y) \geq z \geq -p(C_nG_y)$, and $1-p(C_nG_n) \geq w \geq -p(C_nG_n)$. The set of context effects that satisfy the quantum question (QQ) equality forms a triangular plane that lies inside this pyramid defined by adding one more constraint $q = (y+z) = -(x+w) = 0$. Therefore, the set of context effects that satisfy the QQ equality is a very tiny proportion of the set of all possible context effects.

By our definitions of context effects x, y, z, w above, we have $p(C_yG_n)=p(G_nC_y)-y \geq 0 \rightarrow p(C_yG_n) \geq -y$, and $p(C_nG_y)=p(G_yC_n)-z \geq 0 \rightarrow p(C_nG_y) \geq -z$; so, $p(C_yG_n)+p(C_nG_y) \geq -(y+z) = -q$. Similarly, we also have $p(C_yG_y)=p(G_yC_y)-x \geq 0 \rightarrow p(C_yG_y) \geq -x$, and $p(C_nG_n)=p(G_nC_n)-w \geq 0 \rightarrow p(C_nG_n) \geq -w$; so, $p(C_yG_y)+p(C_nG_n) \geq -(x+w)=q$. Therefore, we obtain the bound $-[p(C_yG_n)+p(C_nG_y)] \leq q \leq [p(C_yG_y)+p(C_nG_n)]$. This bound can be much larger than the possible q value, and we can obtain a tighter bound as described next. We do not use this large bound in the manuscript, and instead use the tighter bound described next.

Order effects are often defined by the difference marginal proportions produced by each order. In particular, define $Cy=[p(G_yC_y)+p(G_nC_y)]-[p(C_yG_y)+p(C_yG_n)]=x+y$ as the order effect for yes to Clinton, and define $Gy=[p(C_yG_y)+p(C_nG_y)]-[p(G_yC_y)+p(G_yC_n)]=-(x+z)$ as the order effect for yes to Gore. Likewise, $Cn=[p(G_nC_n)+p(G_yC_n)]-[p(C_nG_n)+p(C_nG_y)]=(w+z)$ is the order effect for no to Clinton and $Gn=[p(C_nG_n)+p(C_yG_n)]-[p(G_nC_n)+p(G_nC_y)]=-(w+y)$ is the order effect for no to Gore. We could use the sum $Cy+Gy=y-z$ to define the total order effect. In general, however, the answers are arbitrary labels, and thus we could just as well take the sum $Cy+Gn=x-w$. This provides two different ways to define the order effect. Summing in each of these ways is problematic, because each of these sums could be zero, which would misrepresent the fact that there is indeed an order effect. Hence, to measure the order effect size, we eliminate the sign. This produces two order effect sizes, $|z|+|y|$ and $|w|+|x|$, but one could be small or even zero, and the other could be much larger. Therefore, we define the size of the order effect as the maximum of $|z|+|y|$ versus $|w|+|x|$. This also provides a meaningful comparison with the size of our q values. On the one hand, if $|z|+|y|$ is maximum, we can compare this to the magnitude $|q|=|z+y|$, which must less than or equal; if the order effect $|w|+|x|$ is maximum, we can compare this to $|-q|=|w+x|$, which again can be less than or equal.

QQ Equality, Order Effects, and the Nature of the Questions. According to our quantum model of measurement order effects, the relation between two questions determines the order effect, but not the QQ equality. On the one hand, QQ equality holds regardless the two questions' relation and the initial belief/attitude state. It is in this sense, the q test is an a priori and parameter-free test. See latter section in this *SI Text* for the proof of QQ equality. On the other hand, depending on the question relations and the initial belief/attitude state, the model predicts order effects to vary in their directions and sizes. See ref. 1, table 2, for a summary of the predictions on the order effects.

When two questions are completely unrelated, either people generally do not expect order effects, or if the order effects do occur, they do not find the reasons interesting. Thus, the empirical datasets included in the current study are question pairs that are more or less related (e.g., evaluation on two politicians; religious beliefs and political attitude).

Also see the following discussion from ref. 1 (p. 705) for a more detailed discussion on the nature of the questions when we expect question order effects:

> According to the QQ model, two questions may be treated as either compatible or incompatible measurements, and order effects only occur for the latter type. Two questions are *incompatible* if the projectors for the questions are not defined by a common basis. Inability to form a common basis for different questions is expected to occur when the questions are new or unusual, so that an answer must be constructed online. However, if a person has a great deal of experience with a combination, then the person may have sufficient knowledge to form a compatible representation that can answer both questions using a common basis. See Schulkin (2008) for further discussions of this type of cognitive adaption with experience. Therefore, order effects are expected to occur for pairs of incompatible questions, especially uncommon pairs, which must be (partially) constructed on the spot to answer them. Our constructionist view of belief, attitude, and intention is consistent with many others (e.g., Feldman & Lynch, 1988; Schwarz, 2007). From this view, because of cognitive economy, belief, attitude, and intention are not stored in memory as properties, but instead, are constructed while being needed. The retrieved information for one question affects the context for the construction process and influences the subsequent response. Many researchers have investigated the effects of measurements themselves on measured cognition, such as mere measurement effects, self-generated validity theory, and reasons theory (Chandon et al., 2005; Dholakia & Morwitz, 2002; Fitzsimons & Morwitz, 1996; Feldman & Lynch, 1988; Janiszewski & Chandon, 2007). The proposed QQ model attempts to specify and formalize these theories.

Description of the Datasets. The 72 datasets included in the current analysis are 70 field experiments (national surveys) and two laboratory experiments. All of the national surveys recruited national representative samples in the United States, and the laboratory experiments recruited college undergraduate students. In all of the experiments, a randomly selected half of the participants were asked the pair of questions, A and B, in the order of A-then-B. The other half answered exactly the same questions but in the opposite order: B-then-A. The two questions were adjacent to each other. The questions provided binary choices. The datasets are described below.

i) Moore (2) reviewed four types of questions order effects and provided an example dataset from Gallup polls for each type. We obtained the data from him. One of the datasets reviewed by him is excluded from the current analysis because the two questions were not asked back to back, and there was additional context information for one of the questions. According to our quantum model, the QQ equality was predicted not to hold and indeed it did not. The

remaining three datasets from Moore (2) are included in the current analysis. See Table S2.

ii) The current analysis also included a dataset from ref. 3. Almost none of the published studies on question order effects reported sequential probabilities of the questions (cross-tabulation of the portions of responses to the two questions by question order), but the article by Schuman et al. (3) included them, so we are able to use their data to test the QQ equality. The data were collected through Survey Research Center (SRC) national monthly telephone surveys. See Table S3.

iii) Two datasets were collected from between-subjects laboratory experiments by the authors. Undergraduate volunteers were recruited from a large Midwestern university in the United States. One experiment was designed to replicate the Gallup white/black racial hostility study reported by Moore (2). The findings indeed replicated the results found by Gallup in 1996, with a larger effect size. The other experiment was designed to replicate a survey field experiment conducted by Wilson et al. (4). See Table S4.

iv) Finally, we obtained from the Pew Research Center all its surveys that included question order manipulation between two questions during 2001–2011 (10 y) whenever the data are available. See Table S5.

The entire dataset used in the current paper, which contains all of the data from the 72 studies as described, is available in a single file called QuestOrdData.txt. The data file is available from the authors by an e-mail request. The file contains a matrix of 72 rows and 8 columns. The eight columns give the frequencies for AyBy, AyBn, AnBy, AnBn, ByAy, ByAn, BnAy, and BnAn, respectively. Each row is a study. The rows are the following studies, respectively: the 66 Pew studies on various topics, the 3 Gallup studies (Clinton–Gore, Dole–Gingrich, black–white), the two laboratory experiments (black–white, affirmative action), and finally, the abortion questions, as described above.

χ^2 Tests Used in Fig. 1 of Article. First, we present the χ^2 test for order effects. Define n_{YN} as the frequency of saying "yes" to question C when C was asked first and saying "no" to question G when G was asked second, and the other combinations of answers are defined similarly. Define $n = n_{YY} + n_{YN} + n_{NY} + n_{NN}$. Define m_{YN} as the frequency of "yes" to question G when G was asked first and "no" to question C when C was asked second, and the other combinations of answers are defined similarly. Define $m = m_{YY} + m_{YN} + m_{NY} + m_{NN}$. The log-likelihood for the unconstrained model that allows order effects is defined by the following:

$$\begin{aligned} G_U = \big[& n_{YY} \cdot \ln(n_{YY}/n) + n_{YN} \cdot \ln(n_{YN}/n) + n_{NY} \cdot \ln(n_{NY}/n) \\ & + n_{NN} \cdot \ln(n_{NN}/n) + m_{YY} \cdot \ln(m_{YY}/m) + m_{YN} \cdot \ln(m_{YN}/m) \\ & + m_{NY} \cdot \ln(m_{NY}/m) + m_{NN} \cdot \ln(m_{NN}/m) \big]. \end{aligned} \quad \textbf{[S1a]}$$

The log-likelihood for the constrained model that assumes no order effects is defined by the following:

$$\begin{aligned} G_C = \big[& (n_{YY} + m_{YY}) \cdot \ln\big((n_{YY} + m_{YY})/(n+m)\big) \\ & + (n_{YN} + m_{NY}) \cdot \ln\big((n_{YN} + m_{NY})/(n+m)\big) \\ & + (n_{NY} + m_{YN}) \cdot \ln\big((n_{NY} + m_{YN})/(n+m)\big) \\ & + (n_{NN} + m_{NN}) \cdot \ln\big((n_{NN} + m_{NN})/(n+m)\big) \big]. \end{aligned} \quad \textbf{[S1b]}$$

The χ^2 statistic is defined by the difference $\chi^2 = -2 \cdot (G_C - G_U)$. The unconstrained model involves $(4 - 1) + (4 - 1) = 6$ free parameters and the constrained model involves $4 - 1 = 3$ free parameters, and so the χ^2 statistic has df = 3.

Next, we define the χ^2 test for the QQ equality. The log-likelihood for the unconstrained model is defined as follows:

$$\begin{aligned} G_U = \big[& (n_{YN} + n_{NY}) \cdot \ln\big((n_{YN} + n_{NY})/n\big) \\ & + (n_{YY} + n_{NN}) \cdot \ln\big((n_{YY} + n_{NN})/n\big) \\ & + (m_{YN} + m_{NY}) \cdot \ln\big((m_{YN} + m_{NY})/m\big) \\ & + (m_{YY} + m_{NN}) \cdot \ln\big((m_{YY} + m_{NN})/m\big) \big]. \end{aligned} \quad \textbf{[S2a]}$$

The log-likelihood for the model constrained by the QQ equality equals the following:

$$\begin{aligned} G_C = \big[& (n_{YN} + n_{NY} + m_{YN} + m_{NY}) \\ & \cdot \ln\big((n_{YN} + n_{NY} + m_{YN} + m_{NY})/(n+m)\big) \\ & + (n_{YY} + n_{NN} + m_{YY} + m_{NN}) \\ & \cdot \ln((n_{YY} + n_{NN} + m_{YM} + m_{NN})/(n+m)) \big]. \end{aligned} \quad \textbf{[S2b]}$$

The χ^2 statistic is defined by the difference $\chi^2 = -2 \cdot (G_C - G_U)$. The unconstrained model involves $(2 - 1) + (2 - 1) = 2$ free parameters and the constrained model involves $2 - 1 = 1$ free parameter, and so the χ^2 statistic has df = 1.

χ^2 Tests Used to Test Order Effects and *q* Values in the Article. First, we describe the χ^2 test for order effects. The χ^2 statistic for testing an order effect for each of the 66 Pew datasets (not specifically picked to have order effects) was computed using Eqs. **S1a** and **S1b** defined above, producing 66 observed χ^2 values. If the null hypothesis is correct, these should be distributed according to a central χ^2 distribution with df = 3. Ten categories were constructed by computing the nine equally spaced category bounds. These category bounds divide the expected frequency distribution (under the null hypotheses) into 10 equally likely categories, with an equal expected frequency within each of the 10 categories. Then frequency of the 66 observed χ^2 values were counted for each category. Denote f_i as the observed frequency for category i = 1,10. The log-likelihood for the unconstrained model equals the following:

$$G_U = \sum_i f_i \cdot \ln(f_i/66). \quad \textbf{[S3a]}$$

The log-likelihood for the expected frequencies according to the null hypothesis equals the following:

$$G_C = \sum_i f_i \cdot \ln(6.6/66). \quad \textbf{[S3b]}$$

The χ^2 statistic is defined by the difference $\chi^2 = -2 \cdot (G_C - G_U)$. The unconstrained model involves $10 - 1 = 9$ free parameters and the constrained model has no free parameters, and so the χ^2 statistic has df = 9.

Next, we describe the χ^2 test for the QQ equality. The χ^2 statistic for testing the QQ equality for each of the 66 Pew datasets was computed using Eqs. **S2a** and **S2b** defined above, producing 66 observed χ^2 values. If the null hypothesis is correct, these should be distributed according to a χ^2 distribution with df = 1. Ten categories were constructed by computing the nine equally space category bounds. These category bounds divide the expected frequency distribution (under the null hypotheses) into 10 equally likely categories. Then, frequency of the 66 observed χ^2 values was counted for each category. Denote f_i as the observed frequency for category i = 1, 10. Then Eqs. **S3a** and **S3b** were used to compute the log-likelihoods of the unconstrained and constrained models. Once again, the χ^2 statistic is defined by the

difference $\chi^2 = -2 \cdot (G_C - G_U)$. The unconstrained model involves $10 - 1 = 9$ free parameters and the constrained model has no free parameters, and so the χ^2 statistic has df = 9.

Fig. S1 shows two quantile–quantile plots: Fig. S1, *Upper* is for the order effect and the bottom is for the *q* value. The horizontal axis represents the quantile predicted by the χ^2 distribution under the null hypothesis, and the vertical axis represents the observed quantile for the χ^2 statistic. If the null hypothesis is correct, then the points should all fall on the unit slope line. As can be seen in Fig. S1, *Upper* for order effects, the points fall well above the prediction line, and a χ^2 distribution test produced a significant deviation from the null hypothesis ($p = 0.0004$). As shown by Fig. S1, *Lower* for the *q* values, all of the points fall right along the prediction line, and a χ^2 distribution test indicates no significant deviation from the null hypothesis ($p = 0.4625$).

Alternative Models. Wang and Busemeyer (1) have shown that popular models of cognition based upon classical probability theory, such as Bayesian or Markov models, cannot account for both question order effects and the QQ equality at the same time. Without adding ad hoc constraints, they generally predict violation of the QQ equality and thus fail to account for the reported empirical data that support the QQ equality.

In this section, we further explore alternative explanations for the QQ equality based on a repeat-choice process and an anchoring-adjustment process. [We thank Junyi Dai (Indiana University, Bloomington, IN) and Bill Batchelder (University of California, Irvine, CA), and other colleagues who suggested to us these types of models.] We show that to satisfy the QQ equality, the parameters of these models have to be constrained in such a manner that they make the models fail to account for order effects.

First, define an implicit (unobserved or latent) joint probability distribution over binary questions A, B as $[p(\text{AyBy}), p(\text{AyBn}), p(\text{AnBy}), p(\text{AnBn})]$. For example $p(\text{AyBn})$ is the implicit joint probability of saying yes to A and no to B, and these events commute so that $p(\text{AyBn}) = p(\text{BnAy})$. Based on this joint distribution, we can define the standard marginal probabilities as $p(\text{Ay}) = p(\text{AyBy}) + p(\text{AyBn})$ and $p(\text{By}) = p(\text{AyBy}) + p(\text{AnBy})$. We can also define the standard conditional probabilities as $p(\text{Ay}|\text{By}) = p(\text{AyBy})/p(\text{By})$, $p(\text{Ay}|\text{Bn}) = p(\text{AyBn})/p(\text{Bn})$, $p(\text{By}|\text{Ay}) = p(\text{AyBy})/p(\text{Ay})$, and $p(\text{By}|\text{An}) = p(\text{ByAn})/p(\text{An})$. In the models discussed below, the observed proportions are based on these latent joint probabilities plus a bias that depends on the question order. So far, this model (without any bias) has three parameters and it produces no order effects. Next, we show that modified models, which assume a repeat-choice process or an anchoring-adjustment process, cannot account for both order effects and QQ equality at the same time.

Repeat-choice model. Define the observed probabilities for the A-then-B order as $[p_{AB}(\text{AyBy}), p_{AB}(\text{AyBn}), p_{AB}(\text{AnBy}), p_{AB}(\text{AnBn})]$. This model assumes that there is some probability that you repeat the previous choice, and some probability that you make the normal choice conditioned on the first answer. Suppose a is the probability to make a choice that is dependent on the first choice, and define r as the probability of repeating the previous choice. In addition to the three parameters described above, this model adds two additional parameters to make a total of five parameters.

Then for the A-then-B order we have $p_{AB}(\text{AyBy}) = p(\text{Ay})[a \cdot r + (1-a)p(\text{By}|\text{Ay})]$, $p_{AB}(\text{AyBn}) = p(\text{Ay})[a \cdot (1-r) + (1-a)p(\text{Bn}|\text{Ay})]$, $p_{AB}(\text{AnBy}) = p(\text{An})[a \cdot (1-r) + (1-a)p(\text{By}|\text{An})]$, $p_{AB}(\text{AnBn}) = p(\text{An})[a \cdot r + (1-a)p(\text{Bn}|\text{An})]$. Similar equations are used for the B-then-A order.

The following is relatively easy to prove: the context effect $x = p(\text{AyBy}) - p(\text{ByAy}) = a \cdot r \cdot [p(\text{Ay}) - p(\text{By})]$. This model predicts a correlation equal to 1 between the context effect x and the difference between the two marginal probabilities $p(\text{Ay}) - p(\text{By})$. Fig. S2 shows the scatterplot between the context effect $p(\text{AyBy}) - p(\text{ByAy})$ and the difference between marginal probabilities $p(\text{Ay}) - p(\text{By})$. As can be seen, empirically there is little or no correlation ($r = 0.04$), although the predicted correlation equals 1. In particular, the repeat-choice model predicts that $p(\text{AyBy}) - p(\text{ByAy})$ must be smaller than $[p(\text{Ay}) - p(\text{By})]$. However, we find that 56 of the 72 datasets produce a context effect that is bigger than the difference between marginal probabilities.

There is an additional problem faced by the repeat-choice model. The model can be constrained to satisfy the QQ equality, but it fails to account for the order effects. It predicts that context effects are completely determined by the differences in marginal choice probabilities. We computed the χ^2 statistics for all of the 72 datasets. The distribution of the 72 χ^2 statistics obtained by comparing the constrained repeat-choice model (5 parameters) with the unconstrained model (6 parameters) is significantly different from that expected by a central χ^2 distribution with 1 df (the 72 χ^2 statistics were divided into 10 equally spaced categories). The χ^2 statistics equals 168 (df = 9), $p < 0.0001$.

Anchoring-adjustment model. Define the observed probabilities for the A-then-B order as $[p_{AB}(\text{AyBy}), p_{AB}(\text{AyBn}), p_{AB}(\text{AnBy}), p_{AB}(\text{AnBn})]$. These are constructed from the implicit joint probability plus a bias as follows: $p_{AB}(\text{AyBy}) = p(\text{Ay}) \cdot [p(\text{By}|\text{Ay}) + b_{AB}]$, $p_{AB}(\text{AnBn}) = p(\text{An}) \cdot [p(\text{Bn}|\text{An}) + b_{AB}]$, $p_{AB}(\text{AyBn}) = p(\text{Ay}) \cdot [p(\text{Bn}|\text{Ay}) - b_{AB}]$, $p_{AB}(\text{AnBy}) = p(\text{An}) \cdot [p(\text{By}|\text{An}) - b_{AB}]$. The bias b_{AB} can be positive or negative but it must be constrained so that all of the observed probabilities are in fact probabilities (i.e., real numbers between zero and 1).

Similarly, define the observed probabilities for the B-then-A order as $[p_{BA}(\text{ByAy}), p_{BA}(\text{ByAn}), p_{BA}(\text{BnAy}), p_{BA}(\text{BnAn})]$. These are constructed from the implicit joint probability plus a bias as follows: $p_{BA}(\text{ByAy}) = p(\text{By}) \cdot [p(\text{Ay}|\text{By}) + b_{BA}]$, $p_{BA}(\text{BnAn}) = p(\text{Bn}) \cdot [p(\text{An}|\text{Bn}) + b_{BA}]$, $p_{BA}(\text{ByAn}) = p(\text{By}) \cdot [p(\text{An}|\text{By}) - b_{BA}]$, $p_{BA}(\text{BnAy}) = p(\text{Bn}) \cdot [p(\text{Ay}|\text{Bn}) - b_{BA}]$. Again, the bias can be positive or negative, but the bias b_{BA} must be constrained so that the observed probabilities are in fact probabilities.

If the bias for the A-then-B order differs from the bias for the B-then-A order, then this model does not satisfy the QQ equality. The model with unequal bias has five free parameters (leaving only 1 df to test the theory). However, if we force the two biases to be equal, $b_{BA} = b_{AB}$, then this model does satisfy the QQ equality. This constrained model has four free parameters (leaving 2 df's to test the model).

We can estimate each bias directly from each two by two table of responses (see Fig. 1 in the main article for examples) using the mathematical derivation for each bias from the model: $b_{AB} = [p_{AB}(\text{By}) - p_{BA}(\text{By})]/[p_{AB}(\text{Ay}) - p_{AB}(\text{An})]$ and $b_{BA} = [p_{BA}(\text{Ay}) - p_{AB}(\text{Ay})]/[p_{BA}(\text{By}) - p_{BA}(\text{Bn})]$. Fig. S3 below shows one bias plotted as a function of the other computed from the 72 datasets used in the paper. If the biases are equal, then the data points should all fall along a line with zero intercept and unit slope. As can be seen from Fig. S3 below, the observed correlation between the two biases is almost zero, which does not support the equal bias assumption.

An additional problem faced by the anchoring-adjustment type of models is the following: If the two biases for the two orders are constrained to be equal so that the model satisfies the QQ equality, then the model fails to account for the order effects. We computed the χ^2 statistics for all of the 73 datasets. The distribution of χ^2 statistics from the constrained anchoring-adjustment model is significantly different from the distribution expected under the null hypothesis (the null state that there is no difference between the constrained anchoring-adjustment model and the saturated model). The χ^2 statistics equals 39 (df = 9), $p < 0.0001$.

Proof of the QQ Equality. Here, we briefly introduce the basic axioms of quantum theory and then derive the QQ equality. We use the Dirac bracket notation so that $\langle S|T\rangle$ represents the inner

product between two vectors. According to quantum theory, events are represented as subspaces of a Hilbert space. Corresponding to each event A there is a orthogonal projector $\boldsymbol{P}_A$. The state of a quantum system is represented by a unit length vector S within the Hilbert space [Busemeyer and Bruza (ref. 5, appendix C) use a more general density matrix to represent the state]. The probability of event A equals the squared length of the projection $p(A) = \|\boldsymbol{P}_A S\|^2$. If event A is observed, then the state is updated according to Lüder's rule $S_A = \boldsymbol{P}_A S / \|\boldsymbol{P}_A S\|$.

Define S as the initial state (see ref. 5, appendix C, for proof using density matrix representation of state). Denote the projector for saying yes to question C as $\boldsymbol{P_C}$ and denote $\boldsymbol{P_G}$ as the projector for saying yes to question G. We start by expanding the probability for answering "yes" to question C:

$$\begin{aligned}
\|\boldsymbol{P_C}\cdot S\|^2 &= \|\boldsymbol{P_C}\cdot \boldsymbol{I}\cdot S\|^2 = \|\boldsymbol{P_C}\cdot(\boldsymbol{P_G}+(\boldsymbol{I}-\boldsymbol{P_G}))\cdot S\|^2\\
&= \|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S + \boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})\cdot S\|^2\\
&= \|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S\|^2 + \|\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})\cdot S\|^2\\
&\quad + \langle S|\cdot\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})|S\rangle\\
&\quad + \langle S|(\boldsymbol{I}-\boldsymbol{P_G})\cdot\boldsymbol{P_C}\cdot\boldsymbol{P_C}\cdot\boldsymbol{P_G}|S\rangle\\
&= \|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S\|^2 + \|\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})\cdot S\|^2\\
&\quad + \langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})|S\rangle\\
&\quad + \langle S|\cdot\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})|S\rangle *\\
&= \|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S\|^2 + \|\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})\cdot S\|^2\\
&\quad + 2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})|S\rangle]\\
&= \|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S\|^2 + \|\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})\cdot S\|^2\\
&\quad + 2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})|S\rangle],
\end{aligned}$$

and the latter follows from the idempotent property of projectors. (The symbol x^* used in the above derivation refers to the complex conjugate of x.) Define the total probability to say yes to question C when G was asked first as follows:

$$\begin{aligned}
TP_C &= \|\boldsymbol{P_G}\cdot S\|^2\cdot\|\boldsymbol{P_C}\cdot S_G\|^2 + \|\boldsymbol{P}_{\sim G}\cdot S\|^2\cdot\|\boldsymbol{P_C}\cdot S_{\sim G}\|^2\\
&= \|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S\|^2 + \|\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})\cdot S\|^2.
\end{aligned}$$

An order effect for question C when G was asked first is expressed as follows:

$$I_C = TP_C - \|\boldsymbol{P_C}\cdot S\|^2 = -2\cdot\mathrm{Re}\left[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})|S\rangle\right].$$

Immediately, we see that if $\boldsymbol{P_G}$ and $\boldsymbol{P_C}$ commute so that $\boldsymbol{P_G}\cdot\boldsymbol{P_C} = \boldsymbol{P_C}\cdot\boldsymbol{P_G}$ then $\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G}) = \boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot(\boldsymbol{I}-\boldsymbol{P_G}) = 0$ and we predict no order effect. Thus, noncommuting projectors are a necessary condition for order effects. Now let us reexamine the following:

$$\begin{aligned}
I_C &= TP_C - \|\boldsymbol{P_C}\cdot S\|^2 = -2\cdot\mathrm{Re}\left[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot(\boldsymbol{I}-\boldsymbol{P_G})|S\rangle\right].\\
&= -2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}|S\rangle - \langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}\cdot\boldsymbol{P_G}|S\rangle]\\
&= -2\cdot\mathrm{Re}\left[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}|S\rangle - \|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S\|^2\right]\\
&= -2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}|S\rangle] + 2\cdot\|\boldsymbol{P_C}\cdot\boldsymbol{P_G}\cdot S\|^2\\
&= 2\cdot p(GyCy) - 2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}|S\rangle].
\end{aligned}$$

Similarly, the order effect for question G when C was asked first equals the following:

$$\begin{aligned}
I_G &= TP_G - \|\boldsymbol{P_G}\cdot S\|^2\\
&= 2\cdot p(CyGy) - 2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_C}\cdot\boldsymbol{P_G}|S\rangle].
\end{aligned}$$

These two order effects share the same term, $2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_G}\cdot\boldsymbol{P_C}|S\rangle] = 2\cdot\mathrm{Re}[\langle S|\boldsymbol{P_C}\cdot\boldsymbol{P_G}|S\rangle]$. These two terms must be equal because of the mathematical property of inner products:

$$\begin{aligned}
\langle x|y\rangle &= |\langle x|y\rangle|\cdot\left[\cos(\theta) + i\cdot\sin(\theta)\right]\\
\langle y|x\rangle &= \langle x|y\rangle * = |\langle x|y\rangle|\cdot\left[\cos(\theta) - i\cdot\sin(\theta)\right],\\
\mathrm{Re}\left[\langle x|y\rangle\right] &= |\langle x|y\rangle|\cdot\cos(\theta) = \mathrm{Re}\left[\langle y|x\rangle\right].
\end{aligned}$$

The law of reciprocity requires that the transition probability from state y to x be equal to the transition probability from x to y because the transition probabilities are determined by the inner products: $p(x|y) = |\langle x|y\rangle|^2 = |\langle y|x\rangle|^2 = p(y|x)$.

The two order effects, I_C and I_G, together imply the following relation:

$$\begin{aligned}
0 &= \left(2\cdot p(GyCy) - I_C\right) - \left(2\cdot p(CyGy) - I_G\right)\\
&= \left[2\cdot p(GyCy) - p(GyCy) - p(GnCy) + p(Cy)\right]\\
&\quad - \left[2\cdot p(CyGy) - p(CyGy) - p(CnGy) + p(Gy)\right]\\
&= p(GyCy) - p(GnCy) + p(CyGy) + p(CyGn)\\
&\quad - p(CyGy) + p(CnGy) - p(GyCy) - p(GyCn)\\
&= \left[p(CyGn) + p(CnGy)\right] - \left[p(GyCn) + p(GnCy)\right] = q.
\end{aligned}$$

As noted above, the critical property for this prediction is the quantum law of reciprocity, which requires the transition amplitude $\langle T|S\rangle$ from $S \to T$ to be the conjugate of the transition amplitude $\langle S|T\rangle$ from $T \to S$ so that $\langle S|T\rangle = \langle T|S\rangle^*$.

Busemeyer and Bruza (ref. 5, appendix C) present a more general proof using a density matrix to represent the state. The density matrix allows for the state to represent a population with individual differences in the pure state S. This equality also appears as one of the axioms in the axiomatic formulation of quantum theory by Niestegge (6).

Partial Revision Models. Quantum theory assumes that, after a first choice is made, the revised belief state is formed from the initial belief state by projecting the initial belief state onto the subspace for an answer to a question and normalizing the projection to form the revised belief state. More generally, maybe the revision can be "partial" in the sense that the revision is formed by moving some proportion of the belief from the initial belief state to the final projection. We examined the QQ equality under this more general, relaxed assumption for the revision using computer simulation. We examine proportions for revisions of 0, 0.3, 0.5, 0.6, 0.7, and 1.0. (Of course, proportions at the extremes of 0 or 1 exactly satisfy the QQ equality.) We examined spaces ranging in dimension from 2 to 10. We examined two completely different types of unitary transformations that relate the subspaces for the answers to the two questions. We examined subspaces ranging from 1 to 5 in dimension (limited by the size of the entire space). We examined 10,000 random initial states for each of the combinations above. All of the simulations produced QQ equalities of size less than 0.015 and most less than 0.01. So the QQ equality is not exactly satisfied but approximately satisfied for this relaxed version of the theory. The MATLAB program used to perform these simulations is provided below:

```
% Partial revision model: revised state moves to position between initial state and normalized projection
clc
clear

N = 10     % number of dimensions for H
QV = [];
a = .6;    % proportion of movement. a=0 -> stay at initial state
b = 1-a;   %                         a=1 -> move to projection
Trep = 10000;    % number of random selections of initial state

for rep = 1:Trep

    S = rand(N,1);         % random initial state
    S = S./sqrt(S'*S);

%       S = ones(N,1);       % fixed initial state uniform
%       S = S./sqrt(N);

%     H1 = [1 -1; 1 1]./sqrt(2);    % Hadamard gate
%     H = kron(H1,H1);
%     U = kron(H1,H);

   H = zeros(N);                    % Hamiltonian
   H(2:N,1:N-1) = diag(ones(N-1,1));
   H = H + H';
   H = H + diag(1*(1:N));
   U = expm(-1i*H);                 % Unitary

   na = 1;                           % dim of A subspace
   PA = diag([ones(na,1); zeros(N-na,1)]);    % projector for A
   PnA = eye(N) - PA;                        % projector for not A

   SA = PA*S; SA = SA./sqrt(SA'*SA);         % normalized projection A
   Sa = a*SA + b*S;                          % revised state
   Sa = Sa./sqrt(Sa'*Sa);                    % re-normalized
   SnA = PnA*S; SnA = SnA./sqrt(SnA'*SnA);    % normalized projection B
   Sna = a*SnA + b*S;
   Sna = Sna./sqrt(Sna'*Sna);
```

```
    nb = 2;                          % dim of B subspace
    % PB = diag([zeros(N-nb,1); ones(nb,1)]);
    PB = diag([zeros(nb,1); ones(N-nb,1)]);    % projector for B in B coord
    PB = U*PB*U';                    % projector for B in A coord
    PnB = eye(N) - PB;
    % Pb  =  a*PB     + b*eye(N);
    % Pnb =  a*PnB    + b*eye(N);
    % Sb = Pb*S; Sb = Sb./sqrt(Sb'*Sb);
    % Snb = Pnb*S; Snb = Snb./sqrt(Snb'*Snb);
    SB = PB*S; SB = SB./sqrt(SB'*SB);
    Sb = a*SB + b*S;
    Sb = Sb./sqrt(Sb'*Sb);
    SnB = PnB*S; SnB = SnB./sqrt(SnB'*SnB);
    Snb = a*SnB + b*S;
    Snb = Snb./sqrt(Snb'*Snb);

    pAtB =  (S'*PA*S)*(Sa'*PB*Sa);          % prob A then B
    pAtnB = (S'*PA*S)*(Sa'*PnB*Sa);
    pnAtB = (S'*PnA*S)*(Sna'*PB*Sna);
    pnAtnB = (S'*PnA*S)*(Sna'*PnB*Sna);
    pAtB+pAtnB+pnAtB+pnAtnB ;

    pBtA =  (S'*PB*S)*(Sb'*PA*Sb);          % prob B then A
    pBtnA = (S'*PB*S)*(Sb'*PnA*Sb);
    pnBtA = (S'*PnB*S)*(Snb'*PA*Snb);
    pnBtnA = (S'*PnB*S)*(Snb'*PnA*Snb);
    pBtA+pBtnA+pnBtA+pnBtnA;

    x = pAtB - pBtA;                 % order effects
    w = pnAtnB - pnBtnA;
    y = pAtnB - pnBtA;
    z = pnAtB - pBtnA;

    Q1 = x+w;                        % Q value
    Q2 = y+z;
    QV = [ QV Q1 ];
end
max(abs(QV))
```


1. Wang Z, Busemeyer JR (2013) A quantum question order model supported by empirical tests of an a priori and precise prediction. *Top Cogn Sci* 5(4):689–710.
2. Moore DW (2002) Measuring new types of question-order effects. *Public Opin Q* 66(1): 80–91.
3. Schuman H, Presser S, Ludwig J (1981) Context effects on survey responses to questions about abortion. *Public Opin Q* 45(2):216–233.
4. Wilson DC, Moore DW, McKay PF, Avery DR (2008) Affirmative action programs for women and minorities. *Public Opin Q* 72(3):514–522.
5. Busemeyer JR, Bruza PD (2012) *Quantum Cognition and Decision* (Cambridge Univ Press, Cambridge, UK).
6. Niestegge G (2008) An approach to quantum mechanics via conditional probabilities. *Found Phys* 38(3):241–256.

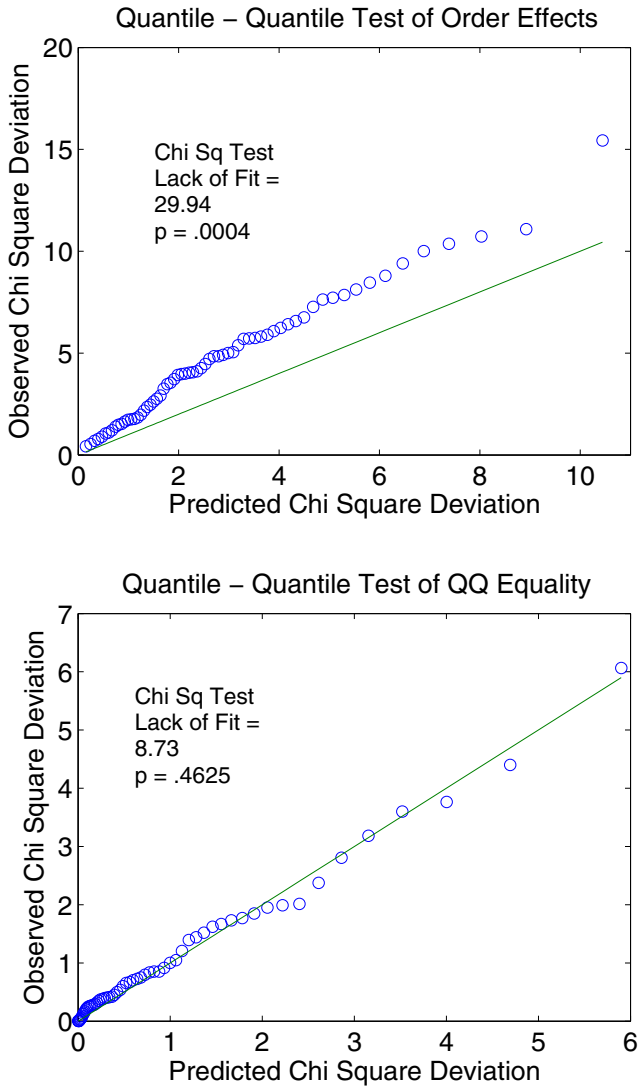


Fig. S1. Quantile–quantile plots of χ^2 statistics. *Upper* shows order effects, and *Lower* shows *q* values. In each panel, the horizontal axis represents the quantile predicted by the χ^2 distribution under the null hypothesis, and the vertical axis represents the observed quantile for the χ^2 statistic. The green line is the prediction for the null hypothesis; the blue circles are the observed values. Deviations indicate departures from the null hypothesis.

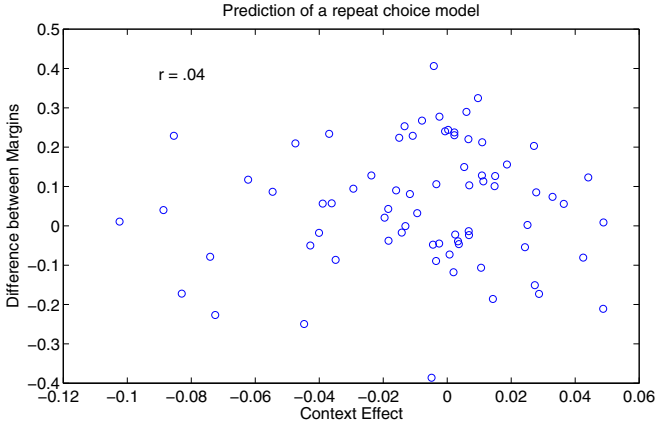


Fig. S2. Correlation between the context effect and the difference between marginal probabilities in the 72 datasets is very small. The blue circles represent the pairs (context effect, difference between marginal probabilities) for each of the 72 studies. The scatter plot shows the relationship between the context effect and the difference between marginal probabilities in each of the 72 studies. The repeat-choice model predicts a correlation equal to 1.

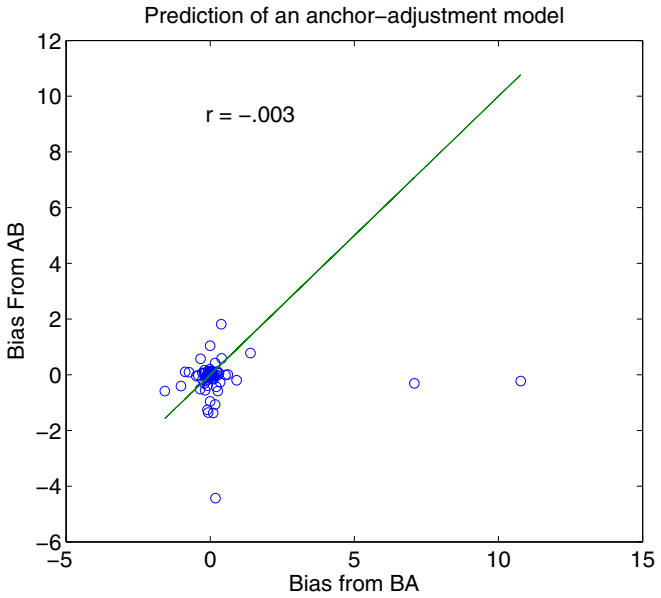


Fig. S3. Correlation between the two biases in the anchoring-adjustment model, as estimated from the 72 datasets, is almost zero. The green line is the regression line with zero intercept and unit slope predicted by the anchor-adjustment model. The blue circles represent the pairs of estimated biases from the AB order and from the BA order (derived from the anchor-adjustment model) obtained from each of the 72 studies. The scatter plot illustrates correlation between the bias in the AB and BA orders. The correlation is predicted to be 1 by the anchor-adjustment model.

Table S1. The context effects in two question sequences

The first order effect	The second order effect
$x = p(G_yC_y) - p(C_yG_y)$	$y = p(G_nC_y) - p(C_yG_n)$
$z = p(G_yC_n) - p(C_nG_y)$	$w = p(G_nC_n) - p(C_nG_n)$

Table S2. Three Gallup field experiments reported by Moore (1)

Gallup surveys	*N*	Question A	Question B
March 27–29, 1995	1,016	Whether you think Newt Gingrich is honest and trustworthy? (Yes/No)	Whether you think Bob Dole is honest and trustworthy? (Yes/No)
June 27–30, 1996	1,005	Do you think that only a few or many white people dislike black people? (A few/Many)	Do you think that only a few or many black people dislike white people? (A few/Many)
September 6–7, 1997	1,003	Do you generally think Bill Clinton is honest and trustworthy? (Yes/No)	Do you generally think Al Gore is honest and trustworthy? (Yes/No)

1. Moore DW (2002) Measuring new types of question-order effects. *Public Opin Q* 66(1):80–91.

Table S3. An SRC field experiments reported by Schuman et al. (1)

Gallup surveys	*N*	Question A	Question B
March–August 1979	651	Do you think it should be possible for a pregnant woman to obtain a legal abortion if she is married and does not want any more children? (Yes/No)	Do you think it should be possible for a pregnant woman to obtain a legal abortion if there is a strong chance of serious defect in the baby? (Yes/No)

1. Schuman H, Presser S, Ludwig J (1981) Context effects on survey responses to questions about abortion. *Public Opin Q* 45(2):216–233.

Table S4. Two laboratory experiments

Lab experiments	*N*	Question A	Question B
2012–2013	116	Do you think that only a few or many white people dislike black people? (A few/Many)	Do you think that only a few or many black people dislike white people? (A few/Many)
2012–2013	118	Do you generally favor or oppose affirmative action (AA) programs for racial minorities? (Yes/No)	Do you generally favor or oppose affirmative action (AA) programs for women? (Yes/No)

Table S5. Sixty-six Pew field experiments during 2001–2011

Pew surveys	*N*	Question A	Question B
2001–2011, 26 studies	Range: 815–3,006 M = 1,644 SD = 422.24	Do you approve or disapprove of the way Bush/Obama is handling his job as President? (Approve/Disapprove)	All in all, are you satisfied or dissatisfied with the way things are going in this country today? (Satisfied/Dissatisfied)
2001–2011, 15 studies	Range: 1,308–1,802 M = 1,504 SD = 95.74	Do you approve or disapprove of the job the Republican leaders in Congress are doing? (Approve/Disapprove)	Do you approve or disapprove of the job the Democratic leaders in Congress are doing? (Approve/Disapprove)
2001–2011, 25 studies	Range: 1,001–2,509 M = 1,612.48 SD = 328.28	Most pair of questions was asked once. Several were asked twice. Questions cover a large variety of topics, from religious beliefs to support to economic policy. For example, a pair of questions asked, "Do you approve or disapprove of the way Barack Obama handled the problems faced by US automakers like General Motors and Chrysler?" and "Do you approve or disapprove of the way Barack Obama handled the problems faced by major US banks and financial institutions?" (Approve/Disapprove) Another example is the following: "In the past presidential election campaign, do you think Barack Obama/John McCain has been too personally critical of John McCain/Barack Obama, or not?" (Too personally critical/Not too personally critical)	

M, mean.